

# SWATHY C

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## PROFILE

Aspiring software developer with hands-on experience building AI-powered applications and solving real-world problems using modern technologies.

## EXPERIENCE

**Prodapt** | Guindy, Chennai | GenAI Developer Intern Dec 2025 – Present

- **Language Translator Widget** — Built in-place translation widget using Python to extract text from audio and text files; integrated **Gemma** and **Azure Translator** backends for multi-format document translation.
- **Invoice Extractor** — Parsed multi-format PDF invoices; mapped extracted fields to **SAP schema** and auto-populated into Excel for client workflows.
- **Talent Match Assistant** — Designed **Neo4j graph DB** schema mapping candidate skills to JDs via Cypher queries; implemented LLM-based interview question generation from resumes.
- **PII Masker Widget** — Combined **Presidio**, **spaCy NER**, and regex to detect and redact PII across document types.
- **ML-Based Guardrails** — Applied **reinforcement learning** and embeddings to detect toxic content; implemented fallback guardrails over LlamaGuard-based pipelines.
- **Real-Time Meeting Interpreter** — Integrated **WebSockets** with **OpenAI** and **Azure Speech**; built **React** frontend; enabled live transcription, web/screen sharing, and meeting bot.
- **Advance Test Case Generator** — Built a **WebSocket**-based live streaming pipeline with **LangChain** to generate classifiable test cases from uploaded docs/user stories, with on-demand test data for exploratory testing.
- **Token Optimizer** — Built an LLM gateway wrapper using an **SLM** to route queries to the lowest-cost capable model, with prompt optimization and response caching to cut token consumption and reduce overall LLM API costs.

**Infosys Springboard** | Virtual | Machine Learning Intern Dec 2025 – Mar 2026

- Built IPL match win prediction using **XGBoost** on historical datasets; feature-engineered run rate, strike rate, and wickets; saved deployment-ready models.

**Altrocks Tech** | Chennai | Software Developer Intern Feb 2024

- Contributed to front-end and back-end tasks using **Java** and **HTML**; gained foundational full-stack web development experience.

**Evolve Robot Lab** | Chennai | Junior Research Fellow Jan – Feb 2024

- Designed and simulated robotic circuits with **Tinkercad** and **Arduino** for real-time embedded control, emphasising sensor integration and microcontroller programming.

## PROJECTS

**Smart IoT Dustbin for Automated Waste Segregation** | *ESP32-CAM, Roboflow, Python, IoT*

- Built a ₹5,000 smart dustbin using ESP32-CAM and Roboflow-trained ML model (92% accuracy); achieved 100% hands-free disposal and ~60% reduction in manual segregation time.

**Defect Classification of Bearings using CNN** | *PyTorch, CNN, Computer Vision*

- Achieved 95%+ accuracy in real-time bearing defect classification; improved predictive maintenance efficiency by 30% and reduced machine downtime by 20%.

**Business Performance Dashboard** | *Power BI, DAX, Data Modeling*

- Unified product, customer, and regional dashboards into a single Power BI executive view; improved decision-making efficiency by 35%.

## EDUCATION

**Rajalakshmi Engineering College** | **B.E Mechatronics Engineering** CGPA: 9.3 | 2022 – 2026

Coursework: Machine Learning, Embedded Systems, Robotics, IoT, Control Systems, Python Programming

**HSC — Velammal Matriculation Hr. Sec. School** 89.66% | 2022

**SSLC — St. Mary's Matriculation Girls Hr. Sec. School** 93.66% | 2020

## SKILLS

**Technical:** Python, Generative AI, LLMs, Prompt Engineering, LangChain, Hugging Face Transformers, FastAPI, REST APIs, Vector Databases, Neo4j, NumPy, Pandas, Git, Power BI

**Soft Skills:** Problem Solving, Teamwork, Time Management

## CERTIFICATIONS

**NPTEL:** Entrepreneurship Essentials, Social Innovation in Industry 4.0, Introduction to IoT, Understanding Incubation and Entrepreneurship, Work System and Design.

**LinkedIn:** Learning Python, Machine Learning with Python, Applied Machine Learning Algorithms, Full Stack Web Development, Java Algorithms, Learning MATLAB.

**TVS Training:** Data Analytics in Microsoft and Power BI.    **Coursera:** AI for Everyone.

## HACKATHONS & ACHIEVEMENTS

- **1st Prize** — 24-Hour Hackathon, VIT Chennai (AR & VR)
- **1st Prize** with Dr. Srinivasan Award — 36-Hour Hackathon "XYNTHRA'25" (Top 33/250+)
- **Best Innovation Award** — 33-Hour Hackathon, Dept. of Mechanical Engg, REC
- **Special Prize** — Women's Hackathon by TANCAM
- **Best Cadet in Semaphore** — NCC Residential Camp
- **3rd Prize** — Project Expo, Devs Club REC
- **Top 45 teams** from 250+ — Smart India Hackathon selection
- Recognised **Top Performer** in department — 6 consecutive semesters
- **2nd Prize** — Dept. IPC, Dr. A.P.J. Abdul Kalam Award
- **Runner-Up** — F-ZONAL Basketball Tournament

## MEMBERSHIP & ROLES

**Content Secretary**, COSMO Club, REC — Leading event content creation and team coordination. (2025 – 2026)

**Class Representative**, REC — Bridge between students and faculty for communication and coordination. (2022 – 2026)

**Content Vice Secretary**, COSMO Club, REC — Assisted in content planning and coordination for events. (2024 – 2025)

**Chairperson**, IEEE RAS, REC — Led robotics events and team planning. (2023 – 2024)